

UNIT – I BASICS OF PYTHON
PART – A

Q.No	Questions	CO	BT	DL
1	What is Python?	CO1	Remember	Easy
2	Define Python Interpreter.	CO1	Remember	Easy
3	List any two features of Python.	CO1	Remember	Easy
4	What are variables in Python?	CO1	Understand	Easy
5	Define data types in Python.	CO1	Understand	Easy
6	What is a list in Python?	CO1	Understand	Medium
7	Differentiate list and tuple.	CO1	Analyze	Medium
8	What is a dictionary?	CO1	Understand	Medium
9	Explain flow control statements.	CO1	Understand	Difficult
10	What is a set in Python?	CO1	Understand	Difficult

PART – B

Q.No	Questions	M	CO	BT	DL
1	Describe the role of the python interpreter in program execution with an example.	13	CO1	Understand	Easy
2	Explain python's language semantics with example of indentation, keyword and naming conventions.	13	CO1	Apply	Easy
3	Discuss python data types in detail with example for each type.	13	CO1	Analyze	Medium
4	Explain in detail the different types of operator in python with examples.	13	CO1	Apply	Medium
5	Write a python program to display all even numbers between 1 and 100 using a loop.	13	CO1	Analyze	Difficult

UNIT – II FUNCTIONS, FILES & OOP

PART – A

Q.No	Questions	CO	BT	DL
1	Define a function in Python.	CO2	Understand	Easy
2	What are function arguments?	CO2	Understand	Easy
3	What is return statement?	CO2	Understand	Easy
4	What is a class in Python?	CO2	Understand	Easy
5	Define object.	CO2	Understand	Easy
6	What is file handling?	CO2	Understand	Medium
7	Differentiate read and write file modes.	CO2	Analyze	Medium
8	What is exception handling?	CO2	Understand	Medium
9	What is import statement?	CO2	Remember	Difficult
10	Define string methods.	CO2	Understand	Difficult

PART – B

Q.No	Questions	M	CO	BT	DL
1	Explain user-defined functions in python with suitable examples.	1 3	CO 2	Apply	Easy
2	Explain how lists can be passed to functions. Write an example program.	1 3	CO 2	Apply	Easy
3	1. Explain constructors and destructors in python with examples.	1 3	CO 2	Analyze	Medium
4	1. Explain file handling in python. Write a program to read and write data into to a text file.	1 3	CO 2	Apply	Medium
5	1. Explain the use of python 'built-in-libraries. Demonstrate math, date time, and random modules with examples.	1 3	CO 2	Apply	Difficult

UNIT – III FOUNDATIONS OF DATA SCIENCE & DESCRIPTIVE ANALYTICS

PART – A

Q.No	Questions	CO	BT	DL
1	Define Data Science.	CO 3	Rememb er	Easy
2	List applications of Data Science.	CO 3	Rememb er	Easy
3	What is data wrangling?	CO 3	Understan d	Easy
4	Define exploratory data analysis.	CO 3	Understan d	Easy
5	What is missing data?	CO 3	Understan d	Easy
6	Define outlier.	CO 3	Understan d	Medium
7	What is descriptive analytics?	CO 3	Understan d	Medium
8	Define mean and median.	CO 3	Understan d	Medium
9	What is standard deviation?	CO 3	Understan d	Difficult
10	Define correlation.	CO 3	Understan d	Difficult

PART – B

Q.No	Questions	M	CO	BT	DL
1	Describe the categories of data for data mining.	1 3	CO 3	Understa nd	Easy
2	Elaborate the application domains of data science.	1 3	CO 3	Apply	Easy
3	Elaborate the different ways to describe or represent data using	1 3	CO 3	Analyze	Medium

	tables with suitable examples.				
4	Describe data warehousing.	1 3	CO 3	Apply	Medium
5	Construct the frequency table and draw bar graph and leaf displays for the following data: 139, 145, 150, 145, 136, 150, 152, 144, 138, 138.	1 3	CO 3	Analyze	Difficult

UNIT – IV NUMPY AND PANDAS**PART – A**

Q.No	Questions	CO	BT	DL
1	Define NumPy.	CO 3	Remember	Easy
2	What is NumPy array?	CO 3	Understand	Easy
3	Define array slicing.	CO 3	Understand	Easy
4	What is Pandas?	CO 3	Remember	Easy
5	Define Series.	CO 3	Understand	Easy
6	Define Data Frame.	CO 3	Understand	Medium
7	What is indexing in Pandas?	CO 3	Understand	Medium
8	What is reindexing?	CO 3	Understand	Medium
9	Define summary statistics.	CO 3	Understand	Difficult
10	What is data alignment?	CO 3	Understand	Difficult

Q.No	Questions	M	CO	BT	DL
1	Explain the creation of arrays in NumPy with examples for different methods such as array(), arrange (), lin space(), and Zeros ().Compare their use cases.	13	CO3	Apply	Easy
2	Explain the concept of subplot sin Matplotlib with an example.	13	CO3	Apply	Easy
3	Discuss indexing, slicing and iterating operations in Numpy arrays. Demonstrate how to access and manipulate elements using these techniques.	13	CO3	Apply	Medium
4	Describe how legends and an notations enhance the readability of a data visualization.	13	CO3	Analyze	Medium
5	Illustrate the significance of choosing appropriate visualization type for different kinds of data.	13	CO3	Analyze	Difficult

UNIT – V DATA VISUALIZATION

PART – A

Q.No	Questions	CO	BT	DL
1	Define data visualization.	CO4	Remember	Easy
2	What is Matplotlib?	CO4	Remember	Easy
3	Define line plot.	CO4	Understand	Easy
4	What is bar plot?	CO4	Understand	Easy
5	Define histogram.	CO4	Understand	Easy
6	What is box plot?	CO4	Understand	Medium
7	What is Sea born?	CO4	Remember	Medium
8	Define scatter plot.	CO4	Understand	Medium

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9	What is pair plot?	CO4	Understand	Difficult
10	What is 3D surface plot?	CO4	Understand	Difficult

PART – B

Q.No	Questions	M	CO	BT	DL
1	Explain Matplotlib plotting techniques.	1 3	CO 4	Apply	Easy
2	Discuss different types of plots with examples.	1 3	CO 4	Apply	Easy
3	Analyze advanced visualization using Seaborn.	1 3	CO 4	Analyze	Medium
4	Explain styling and customization of plots.	1 3	CO 4	Apply	Medium
5	Create meaningful visualizations for real-world datasets.	1 3	CO 4	Evaluate	Difficult